

Seyed Alireza Mousavian Anaraki

PhD in Data Science, University of Rome Tor Vergata | Visiting Researcher, University of Sheffield

Natural Language Processing (NLP) | Artificial Intelligence (AI) | Machine Learning (ML) |
Sustainability (ESG) Reporting | AI for Social Good

Nationality: Iranian

Date of birth: 22/11/1996

🏠 Sheffield, UK

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🔍 [Google Scholar](#)

📄 [Research Gate](#)

🌐 [LinkedIn](#)

EDUCATION

- 01/11/2023-30/10/2026 (Expected completion; thesis defense scheduled for April–May 2027)

PhD student in Data Science at university of Rome Tor Vergata, Rome, Italy

Research Focus: Natural Language Processing (NLP) for sustainability (ESG) analysis, particularly on large-scale processing of sustainability reports, weak-supervision techniques, and the use of large language models (LLM) for structure, consistency, and credibility assessment.

Supervisors: Profs. Danilo Croce and Roberto Basili **Email:** {croce, basili}@dii.uniroma2.it

- 23/09/2019 – 13/11/2021

Master's degree in Industrial Engineering (Engineering Management), Iran University of Science and Technology, Department of Industrial Engineering, Tehran, Iran

legal duration of the course= 2 years, GPA: 19.33/20 (Ranked 1st student in Master GPA)

Thesis Title: Development and implementation of data mining methods by creating strong, balanced, and interpretable clusters to adopt appropriate policies

Supervisors: Dr. Abdorrahman Haeri **Email:** ahaeri@iust.ac.ir [Google Scholar](#)

- 23/09/2015 – 27/08/2019

Bachelor's degree in Industrial Engineering, Kashan University, Department of Industrial Engineering, Kashan, Iran

legal duration of the course= 4 years, GPA: 18.62/20 (Ranked 1st student in Bachelor GPA)

RESEARCH INTERESTS

- ✓ Artificial Intelligence (AI), Deep Learning (DL), and Machine Learning (ML)
- ✓ Information Extraction and Natural Language Processing (NLP) Applications
- ✓ Interpretability and Analysis of NLP Models
- ✓ NLP and Large Language Models (LLMs) for Sustainability (ESG) Reporting
- ✓ AI-Driven Decision-Making in Management Applications
- ✓ Operations Research Applications

TEACHING ASSISTANT

- 23/09/2017 – 19/02/2018

Teacher assistant for **Linear Algebra** course at Kashan university, Kashan, Iran

Supervisor: Dr. Mohammad Taghi Rezvan

Email: rezvan@kashanu.ac.ir

RESEARCH EXPERIENCE

- 01/04/2026 – 14/08/2026

Visiting Researcher at the School of Computer Science, The University of Sheffield, Sheffield, England, United Kingdom.

- ✓ Conducted research on investigating **credibility bias** in **ESG language models**, focusing on how geographic, institutional, and linguistic factors influence **automated sustainability assessment**.

Supervisor: Dr. Nafise Sadat Moosavi

Email: n.s.moosavi@sheffield.ac.uk

- 21/11/2020 – 22/08/2021

Researcher and data analyzer at Iran's National Elites Foundation (INEF), Tehran, Iran.

- ✓ Participated in data analytics research group as **data-driven reliability-centered maintenance specialist** for critical **energy infrastructure maintenance improvement**.

Supervisor: Dr. Alireza Fereidunian

Email: fereidunian@kntu.ac.ir

- 23/09/2019 – 13/11/2021

Research assistant at Iran University of Science and Technology, Tehran, Iran

- ✓ Participated as a research assistant on a team led by Dr. Abdorrahman Haeri to **develop and implement data mining methods in different fields with an emphasis on data-driven decision-making**.

Supervisors: Dr. Abdorrahman Haeri

Email: ahaeri@iust.ac.ir

RESEARCH ACHIEVEMENTS

Journal Papers

- ✓ Anaraki, S. A. M., Croce, D., & Basili, R. (2025). Large language models for sustainability reporting: A systematic review and research agenda. *Sustainable Futures*, 10, 101494. <https://doi.org/10.1016/j.sftr.2025.101494>
- ✓ Fallahi, A., Mousavian Anaraki, S. A., Mokhtari, H., & Niaki, S. T. A. (2022). Blood plasma supply chain planning to respond COVID-19 pandemic: a case study. *Environment, Development and Sustainability*, 1-52. <https://doi.org/10.1007/s10668-022-02793-7>
- ✓ Anaraki, S. A. M., & Haeri, A. (2022). Soft and hard hybrid balanced clustering with innovative qualitative balancing approach. *Information Sciences*. <https://doi.org/10.1016/j.ins.2022.09.044>
- ✓ Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2022). Generating balanced and strong clusters based on balance-constrained clustering approach (strong balance-constrained clustering) for improving ensemble classifier performance. *Neural Computing and Applications*, 1-17. <https://doi.org/10.1007/s00521-022-07595-6>

- ✓ Mousavian Anaraki, S. A., Haeri, A., & Moslehi, F. (2021). A hybrid reciprocal model of PCA and K-means with an innovative approach of considering sub-datasets for the improvement of K-means initialization and step-by-step labeling to create clusters with high interpretability. *Pattern Analysis and Applications*, 24(3), 1387–1402. <https://doi.org/10.1007/s10044-021-00977-x>
- ✓ Mousavian, S. A., Haeri, A., & Moslehi, F. (2021). Providing a Hybrid Clustering Method as an Auxiliary System in Automatic Labeling to Divide Employee into Different Levels of Productivity and their Retention. *Iranian Journal of Management Studies*. <https://doi.org/10.22059/ijms.2021.299705.674004>
- ✓ Rahimi, F., & Mousavian Anaraki, S. A. (2020). Proposing an Innovative Model Based on the Sierpinski Triangle for Forecasting EUR/USD Direction Changes. *Journal of Money and Economy*, 15(4), 423–444. <https://doi.org/10.52547/jme.15.4.423>

Conference Papers

- ✓ Mousavian Anaraki, S. A., Croce, D., Costa, R., Tiburzi, L., Calabrese, A., & Basili, R. (2026). Unsupervised GRI-TCFD Alignment with LLM-Assisted Validation for Climate Disclosure and Greenwashing Risk Analysis . In *Proceedings of the 2nd Workshop on Ecology, Environment, and Natural Language Processing* (pp. 15–25). *European Language Resources Association (ELRA)*. <https://doi.org/10.63317/4pebkuscwua5>.
- ✓ Borazio, F., Anaraki, S. A. M., Rai, S. I., Croce, D., & Basili, R. (2026). **UniTor at PFB: Constrained Agentic Prompting and Self-Consistency for Financial Multi-Choice QA**. In *EVALITA*.
- ✓ Anaraki, S. A. M., Croce, D., & Basili, R. (2025, September). **Automatic GRI-SDG Annotation and LLM-Based Filtering for Sustainability Reports**. In *Proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)* (pp. 775-784).
- ✓ Anaraki, S. A. M., Croce, D., & Basili, R. (2025, July). Unsupervised Sustainability Report Labeling based on the integration of the GRI and SDG standards. In *Proceedings of the Fourth Workshop on NLP for Positive Impact (NLP4PI)* (pp. 151-162). <https://aclanthology.org/2025.nlp4pi-1.13/>

HONORS AND AWARDS

- **July 2025**
Outstanding Paper Award, Workshop on NLP for Positive Impact (NLP4PI @ ACL 2025)
- **July 2020**
Ranked 1st student in Master GPA among near 20 students in the “Department of Industrial Engineering” of Iran University of Science and Technology
- **September 2019**
Eligible to continue my education in the Master’s Program at the Iran University of Science and Technology without taking the “National University Entrance Exam”
- **July 2018**
Ranked 1st student in Bachelor GPA among near 40 students in the “Department of Industrial Engineering” of Kashan university

LANGUAGE SKILLS

- **English:** Upper Intermediate
- **Persian:** Native

COMPUTER SKILLS

- **Programming Software:**

Python, R

Database management system: SQL Server

- **Engineering Software:**

Data visualization/BI tools: Tableau, Power BI

Other: Gams, COMFAR, Expert Choice, ALDEP, VOS viewer